

Checkpoint 0: Choose Your Scenario

 Important

Due 10/18 (4 points). See [Project Central](#) for the full project overview.

Goal

Choose the health scenario you will follow all quarter, and gather the real-world numbers you will need for Checkpoints 1–3.

Tasks

1. Team and scenario

- State whether you are working alone or with a partner (and your partner's name).
- Choose a scenario from the [scenario menu](#) **or** propose your own.
- In 3–5 sentences, describe the scenario: Who is the population? What is being tested, counted, or measured? Why does it matter to a clinician or public health decision-maker?

Proposing your own scenario?

Your scenario needs three pieces:

1. a **test or classification** with a sensitivity/specificity (or two conditional probabilities) for Checkpoint 1,
2. a **count** of events that could be modeled with a discrete distribution for Checkpoint 2, and
3. a **time or measurement** that could be modeled with a continuous distribution for Checkpoint 3.

If you are unsure, email me or come to office hours before the due date.

2. Find your input values

Fill in the table below. For each quantity, find a published value and cite it. You will need **at least one value for each checkpoint**. Values can be approximate (e.g., “about 1 in 1,000”).

Checkpoint	Quantity	Value	Source
1	Prevalence (or base rate)		
1	Sensitivity (or $P(+ \mid \text{condition})$)		
1	Specificity (or $P(- \mid \text{no condition})$)		
2	Rate or probability for your count		
3	Mean (and SD, if available) of your time/measurement		

Acceptable sources include CDC or WHO reports, USPSTF evidence reviews, government health statistics, and peer-reviewed articles. Wikipedia is fine for finding a source, but cite the original source.

3. Your guiding questions

Write one question for each checkpoint that your analysis will answer. For example (emergency department scenario):

- **CP 1:** If a patient is triaged at level 2, what is the probability they are admitted?
- **CP 2:** What is the probability that more than 15 patients arrive in one hour?
- **CP 3:** What is the probability that the average wait time of the next 50 patients exceeds 3 hours?

These can change later. This step is to make sure your scenario will work for all three checkpoints.

Submission

Submit a rendered HTML file (and your `.qmd`) with the three sections above. One submission per team.

Rubric (4 points)

Criterion	1 point	0 points
Scenario is described clearly and fits all three checkpoints	Yes	Missing or does not fit
Input table has at least one value for each checkpoint	Yes	Incomplete
Every value has a credible, cited source	Yes	Missing or unreliable sources
One guiding question for each checkpoint	Yes	Missing